

Finding Sign and Like-Term Errors in Polynomial Work

Distributing a minus sign, combining only like terms, and checking a simplification with a test value

Directions

- Four of these problems show somebody else's work. For each one, find the mistake, check the two expressions against a test value, and write the corrected expression.
 - A test value is a quick way to catch an error: if two expressions really are equivalent, they agree at *every* number you substitute. One disagreement proves a mistake.
 - Write every corrected expression in standard form.
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1. Simplify: $(4x + 7) - (x - 3)$

Answer: _____

2. Simplify: $5x^2 + 3x - 2x^2 + x$

Answer: _____

3. Dev simplified $(5x^2 - 3x) - (2x^2 + 4x)$ and wrote $3x^2 + x$.

(a) Name the error in Dev's work and say which step it happened at.

Answer: _____

(b) Substitute $x = 2$ into the *original* expression to find its true value.

Answer: _____

(c) Write the corrected simplified expression.

Answer: _____

4. Simplify: $6x^3 - 2x^2 + x^3 + 5x^2$

Answer: _____

5. Mara simplified $3x^2 + 4x + 2x^2$ and wrote $9x^3$.

- (a) Name the error in Mara's work. There are two things wrong with the single step she took — describe both if you can.

Answer: _____

- (b) Substitute $x = 2$ into the *original* expression to find its true value.

Answer: _____

- (c) Write the corrected simplified expression.

Answer: _____

6. Theo expanded $(2x - 5)(x - 3)$ and wrote $2x^2 - 11x - 15$.

(a) Only one term of Theo's answer is wrong. Name the error that produced it.

Answer: _____

(b) Substitute $x = 2$ into the *original* product to find its true value.

Answer: _____

(c) Write the corrected expansion.

Answer: _____

7. Simplify, working left to right and taking care with each bracket:

$$(3x^2 - 5x + 2) - (x^2 - 5x - 6) + (2x - 1)$$

Answer: _____

8. A student simplified $(x + 4)^2 - (x - 2)(x + 2)$ like this:

$$(x + 4)^2 - (x - 2)(x + 2) = x^2 + 16 - x^2 - 4 = 12$$

- (a) There are two separate errors in that line. Name both.

Answer: _____

- (b) Substitute $x = 3$ into the *original* expression to find its true value.

Answer: _____

- (c) Write the corrected simplified expression.

Answer: _____